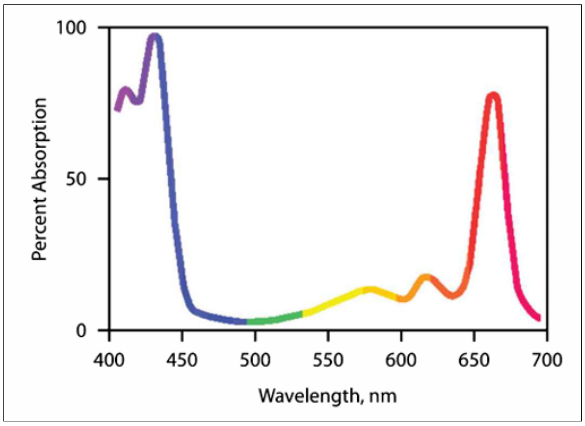
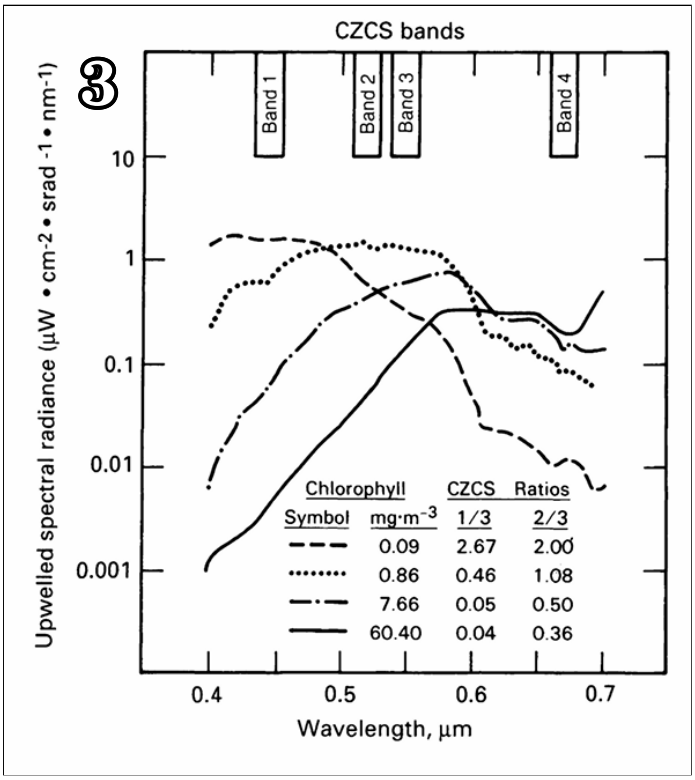
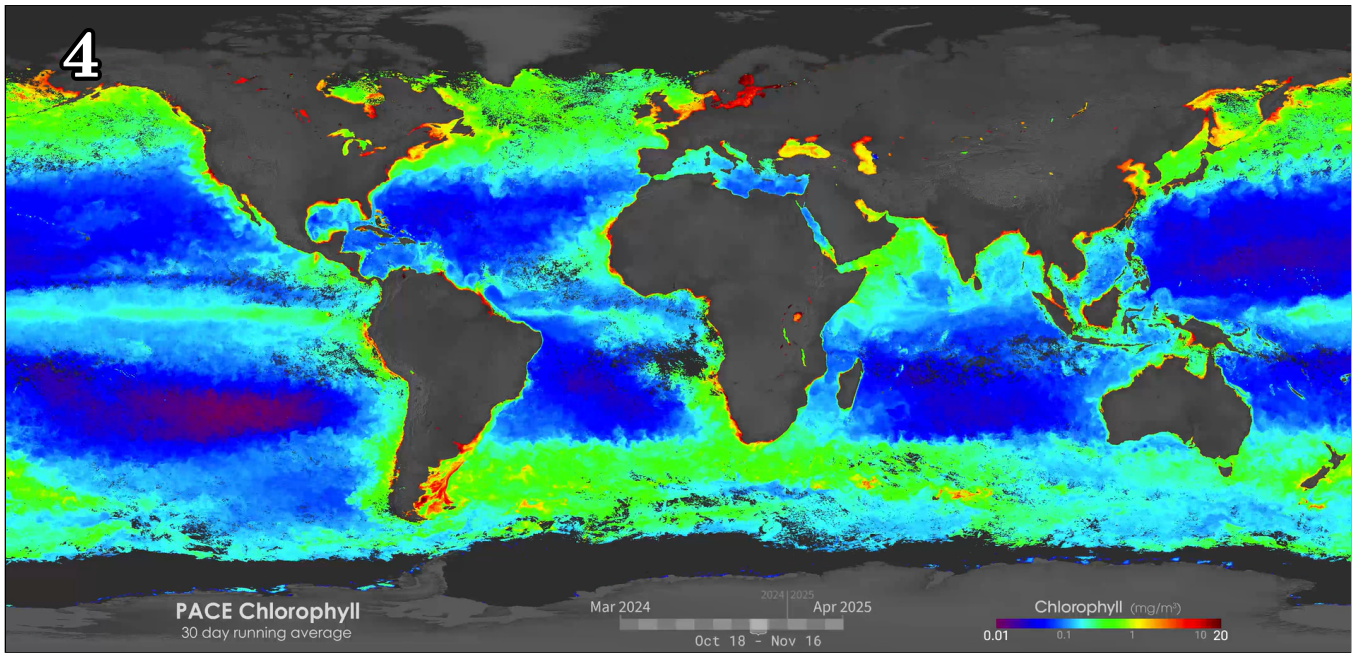
2


100
50
0
Percent Absorption
400
450
500
550
600
650
700
Wavelength, nm

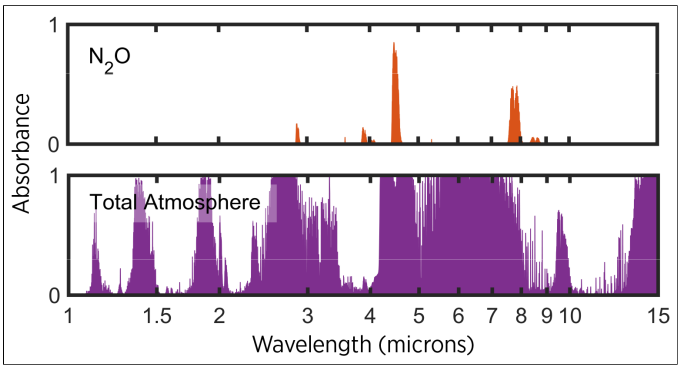
3


CZCS bands
Band 1
Band 2
Band 3
Band 4
Upwelled spectral radiance (µW • cm-2 • srad -1 • nm-1)
10
1
0.1
0.01
0.001
Chlorophyll
Symbol
mg·m-3
CZCS Ratios
1/3
2/3
0.09
2.67
2.00
0.86
0.46
1.08
7.66
0.05
0.50
60.40
0.04
0.36
0.4
0.5
0.6
0.7
Wavelength, µm

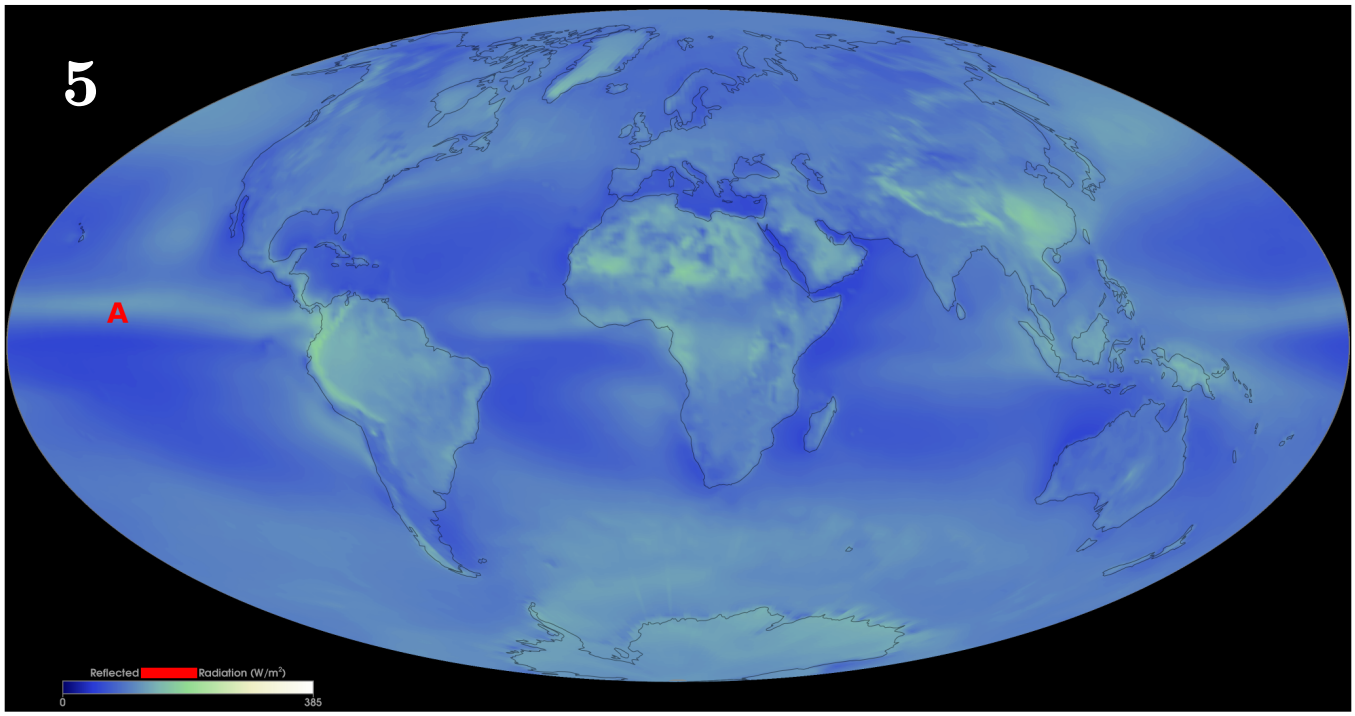
4


PACE Chlorophyll
30 day running average
Mar 2024
2024 2025
Apr 2025
Oct 18 - Nov 16
Chlorophyll (mg/m³)
0.01
0.1
1
10
20

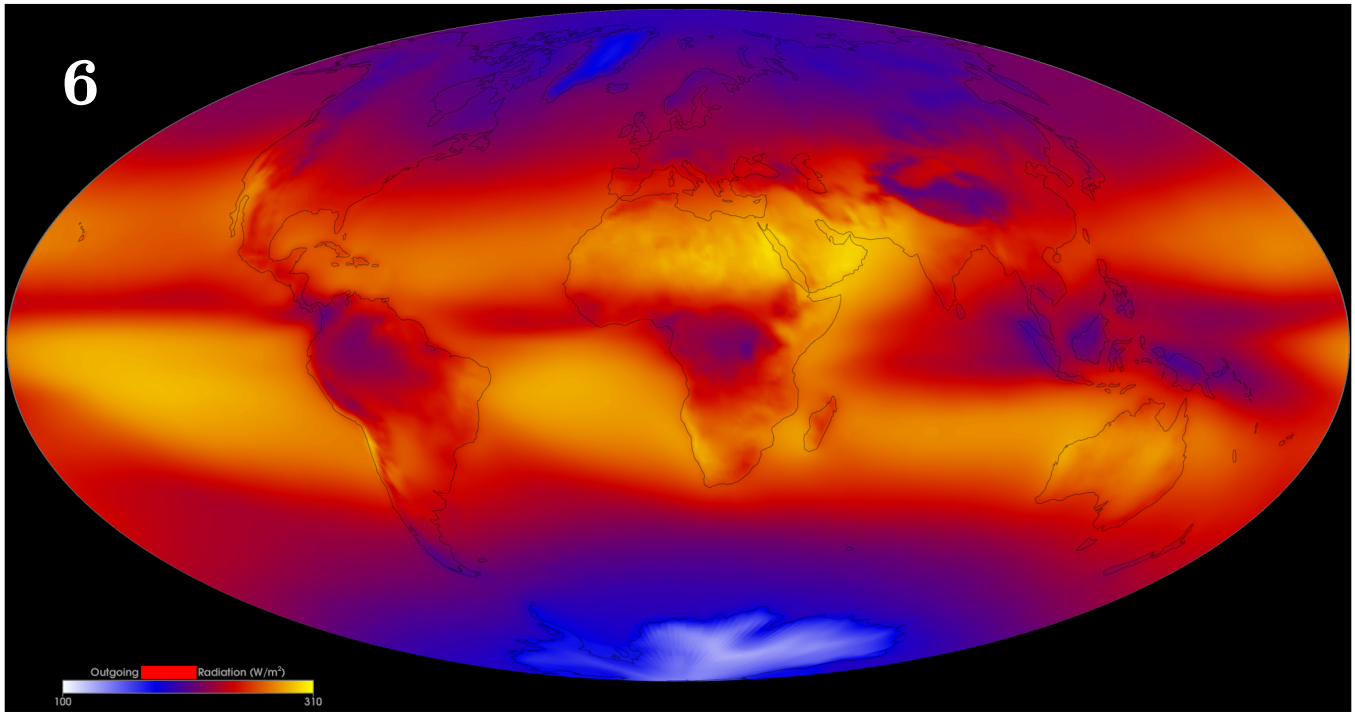
1


N_2O
Total Atmosphere
Absorbance
1
0
1
0
1
1.5
2
3
4
5
6
7
8
9
10
15
Wavelength (microns)

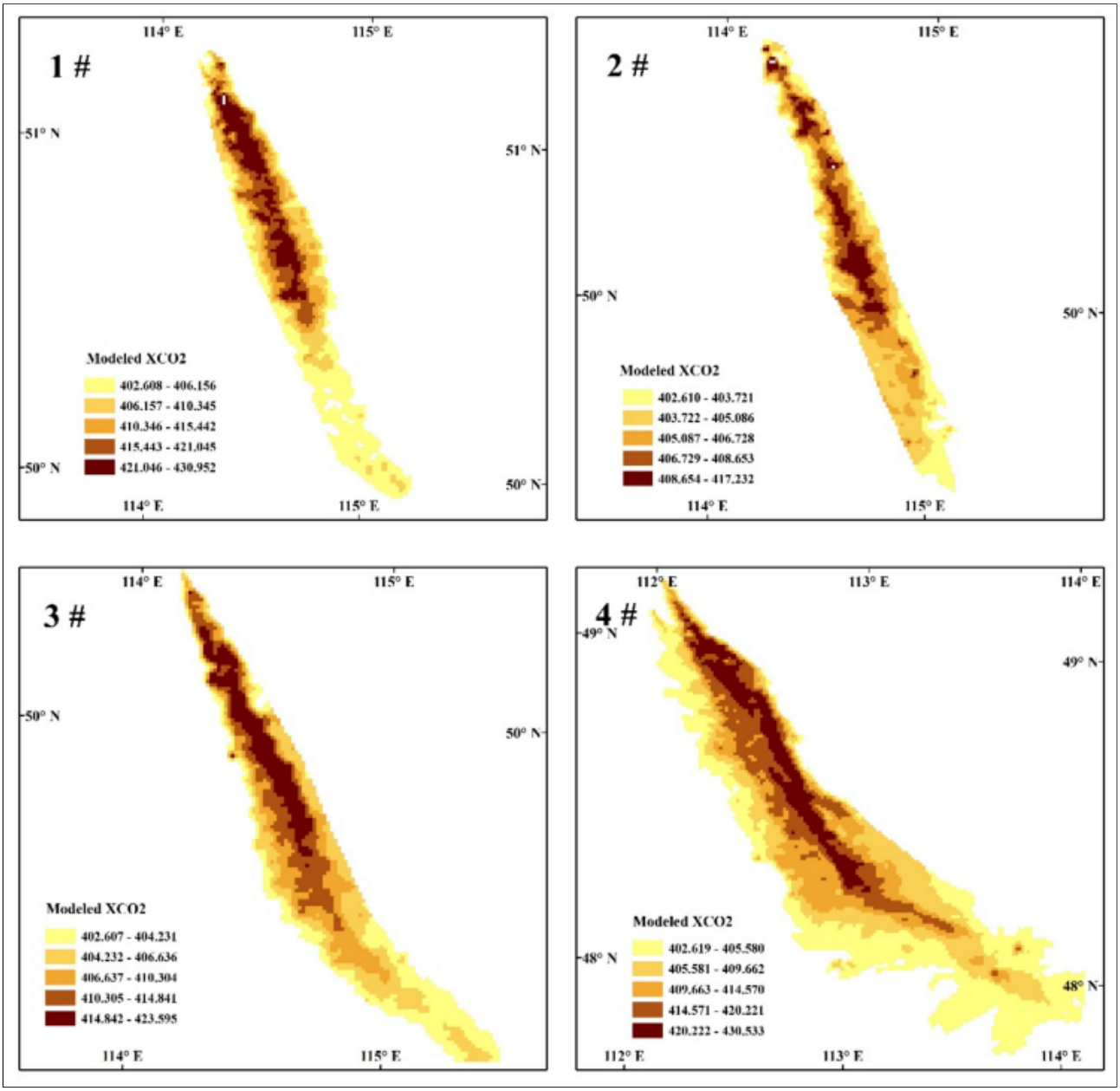
5


A
Reflected Radiation (W/m²)
0
385

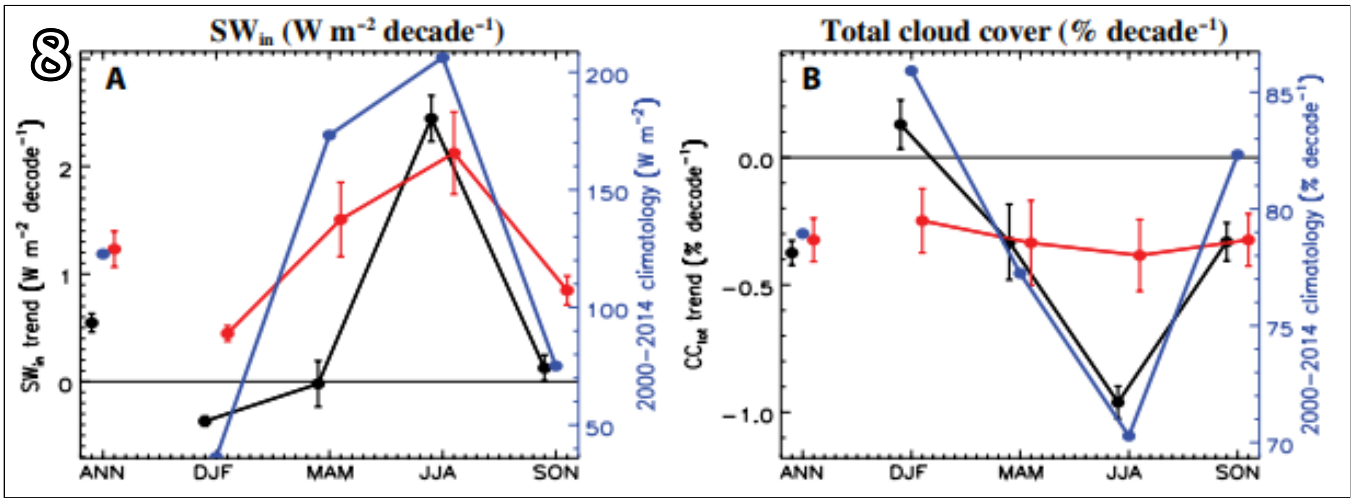
6

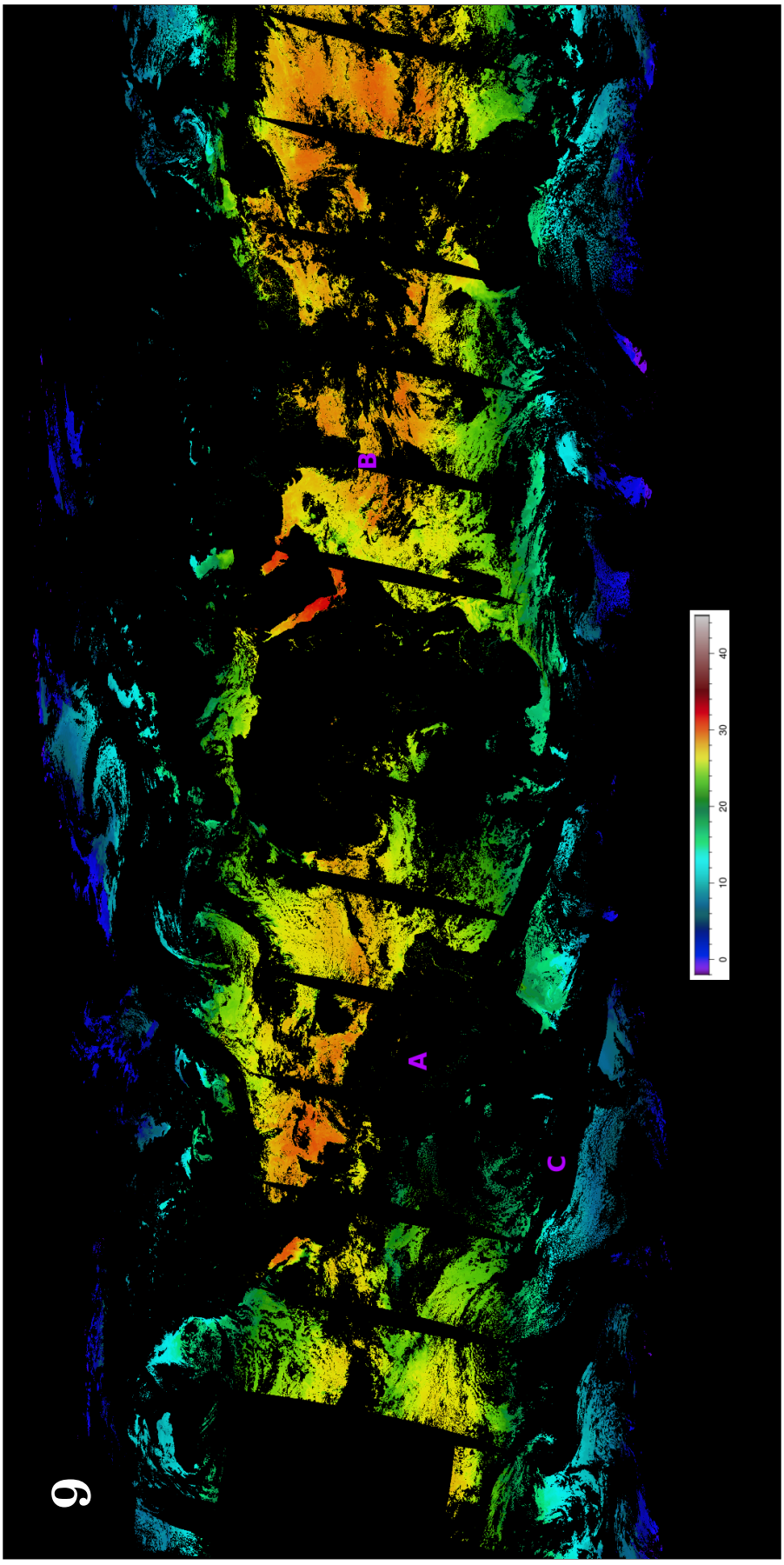

Outgoing Radiation (W/m²)
100
310

7

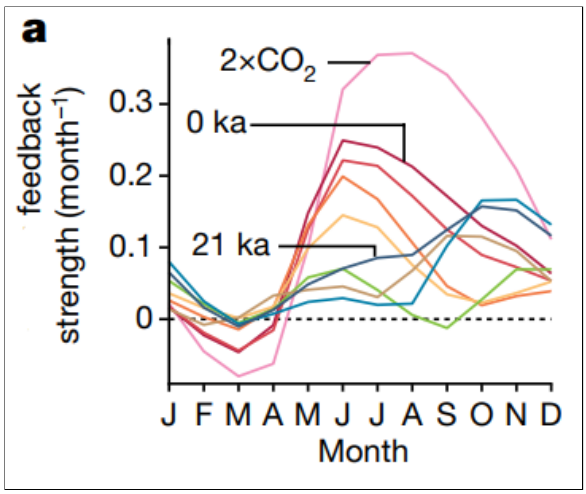

1 #
114° E
115° E
51° N
50° N
Modeled XCO2
402.608 - 406.156
406.157 - 410.345
410.346 - 415.442
415.443 - 421.045
421.046 - 430.952
2 #
Modeled XCO2
402.610 - 403.721
403.722 - 405.086
405.087 - 406.728
406.729 - 408.653
408.654 - 417.232
3 #
Modeled XCO2
402.607 - 404.231
404.232 - 406.636
406.637 - 410.304
410.305 - 414.841
414.842 - 423.595
4 #
112° E
113° E
114° E
49° N
48° N
Modeled XCO2
402.619 - 405.580
405.581 - 409.662
409.663 - 414.570
414.571 - 420.221
420.222 - 430.533

8

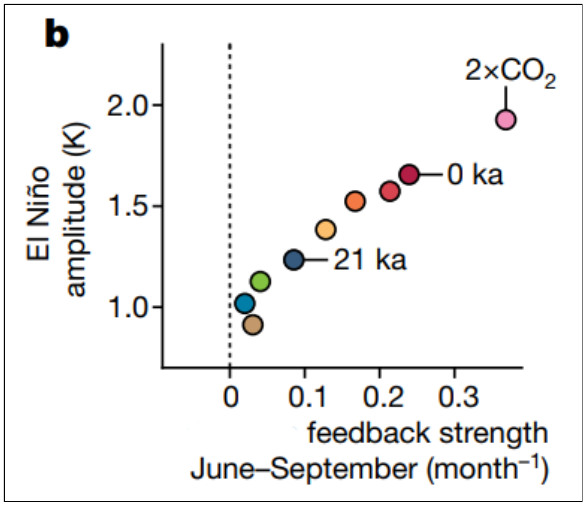

SW$_{in}$ (W m^{-2} decade^{-1})
A
SW$_{in}$ trend (W m^{-2} decade^{-1})
2000–2014 climatology (W m^{-2})
ANN
DJF
MAM
JJA
SON
Total cloud cover (% decade^{-1})
B
CC$_{tot}$ trend (% decade^{-1})
2000–2014 climatology (% decade^{-1})


9
A
B
C
0
10
20
30
40

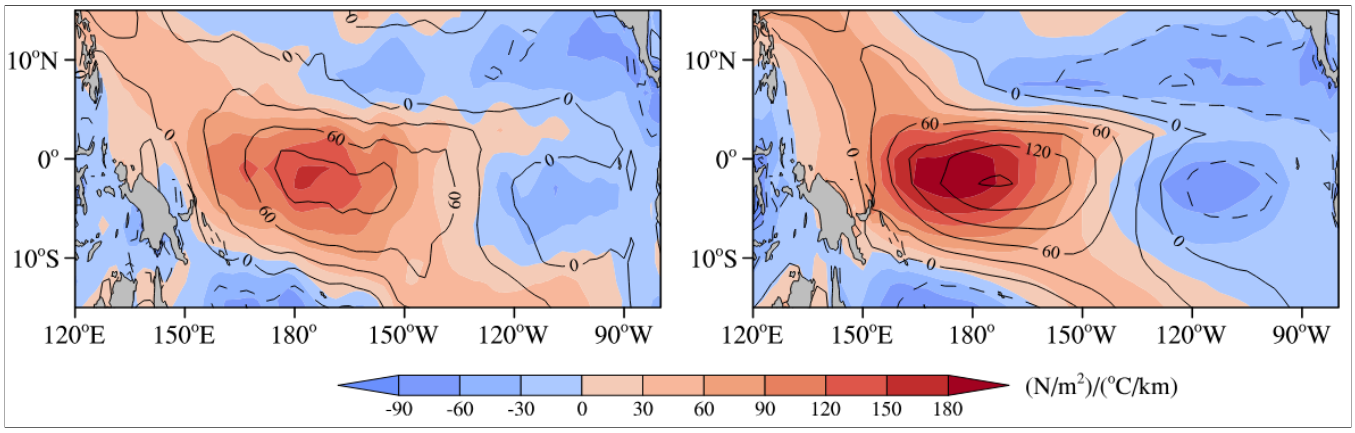
10


a
feedback strength (month⁻¹)
0.3
0.2
0.1
0
2×CO2
0 ka
21 ka
J F M A M J J A S O N D
Month

11


b
El Niño amplitude (K)
2.0
1.5
1.0
2×CO2
0 ka
21 ka
0
0.1
0.2
0.3
feedback strength
June–September (month⁻¹)

12


10°N
0°
10°S
120°E
150°E
180°
150°W
120°W
90°W
(N/m²)/(°C/km)
-90
-60
-30
0
30
60
90
120
150
180